UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA

Scholarship with Professionalism

The Four Dimensions of Service Management

ITIL® 4 Foundation | Chapter 3

Lecture slides prepared for undergraduate and postgraduate students of Information Systems and IT Service Management

Presented by: Prof. Isaac Kofi Mensah • Department of Information Technology Studies

ORIENTATION

About This Session

1 Source text

AXELOS (2019), ITIL® Foundation: ITIL 4 Edition, Chapter 3. All content is summarised and re-expressed for teaching purposes.

2 Level & duration

Foundation level. Approximately two contact hours plus one hour of guided tutorial activity.

3 Examinable weight

The four dimensions carry a defined share of the ITIL 4 Foundation examination. Expect recall and simple application questions.

4 How we will learn

Concept exposition, the Axle Car Hire running illustration, Ghanaian case applications, group activity and practice questions.

5 Prerequisites

Chapter 1 (Introduction) and Chapter 2 (Key concepts of service management), especially value, outcomes, costs and risks.

6 Assessment link

Group assignment: apply the four dimensions to a Ghanaian public or private sector digital service of your choice.

BY THE END OF THIS SESSION YOU WILL BE ABLE TO

Learning Objectives (1 of 2)

Knowledge and comprehension

These first objectives address the recall and understanding levels of Bloom's taxonomy. They map directly to the ITIL 4 Foundation syllabus statement on describing the four dimensions of service management.

1 Explain the purpose of the four dimensions

Describe why a holistic perspective is necessary and what happens when organisations focus on one dimension alone.

2 Name and define each dimension

Organisations and people; information and technology; partners and suppliers; value streams and processes.

3 Describe the scope of each dimension

State what each dimension covers both for the Service Value System as a whole and for an individual service.

4 Recognise the external factors

Identify the PESTLE factors and explain how they constrain or influence a service provider.

BY THE END OF THIS SESSION YOU WILL BE ABLE TO

Learning Objectives (2 of 2)

Application and analysis

These objectives go beyond the examination. They prepare you to diagnose real service problems in Ghanaian organisations and to argue for balanced investment across all four dimensions.

5 Apply the dimensions to a real service

Analyse a live digital service and classify its components under the correct dimension.

6 Diagnose imbalance

Explain how neglect of one dimension produces failure symptoms in the others, using evidence from a case.

7 Evaluate contemporary developments

Assess how cloud computing, artificial intelligence, data protection law and sourcing trends reshape each dimension.

8 Distinguish value streams from processes

Compare the two concepts and explain why the distinction matters for improvement work.

AGENDA

Chapter Roadmap

1

Why four dimensions?

The holistic argument

2

Organisations and people

Structure, culture, competence

3

Information and technology

Data, systems, cloud

4

Partners and suppliers

Sourcing and integration

5

Value streams and processes

Activities and workflow

6

External factors

The PESTLE lens

7

Contemporary issues

AI, sovereignty, resilience

8

Practice and revision

Activity and exam questions

RECAP OF CHAPTERS 1 AND 2

Where We Are Coming From

1 The purpose of an organisation

An organisation exists to create value for its stakeholders. Value is co-created in a relationship, never handed over in one direction.

2 Value comes through services

Value is realised through the provision and the consumption of services, supported by products and resources.

3 The Service Value System

The SVS explains how the components and activities of an organisation combine to create value. Chapter 4 examines it in full.

4 Why the dimensions come first

The four dimensions apply to every element of the SVS, so they must be understood before the SVS itself is unpacked.

The Problem: Organisational Tunnel Vision

THE CORE ARGUMENT

Organisations frequently concentrate their attention on one area of an initiative and let the others drift. No single aspect of service management is sufficient on its own to deliver the outcomes an organisation needs.

Process without people

A workflow is redesigned on paper, but nobody consults the staff, partners or technologies that must carry it out. Adoption collapses.

Technology without process

A platform is procured and installed with no regard for the procedures or the people it was meant to support. The tool becomes shelfware.

Structure without culture

An organisation chart is redrawn while the shared values and attitudes that actually govern behaviour remain untouched.

DEFINITION

Key Message: The Four Dimensions

KEY MESSAGE

To support a holistic approach, ITIL defines four dimensions that are collectively critical to the effective and efficient facilitation of value for customers and other stakeholders in the form of products and services.

1 Organisations and people

Structures, roles, responsibilities, authority, culture, staffing and competencies.

2 Information and technology

The information and knowledge required, and the technologies that support and enable services.

3 Partners and suppliers

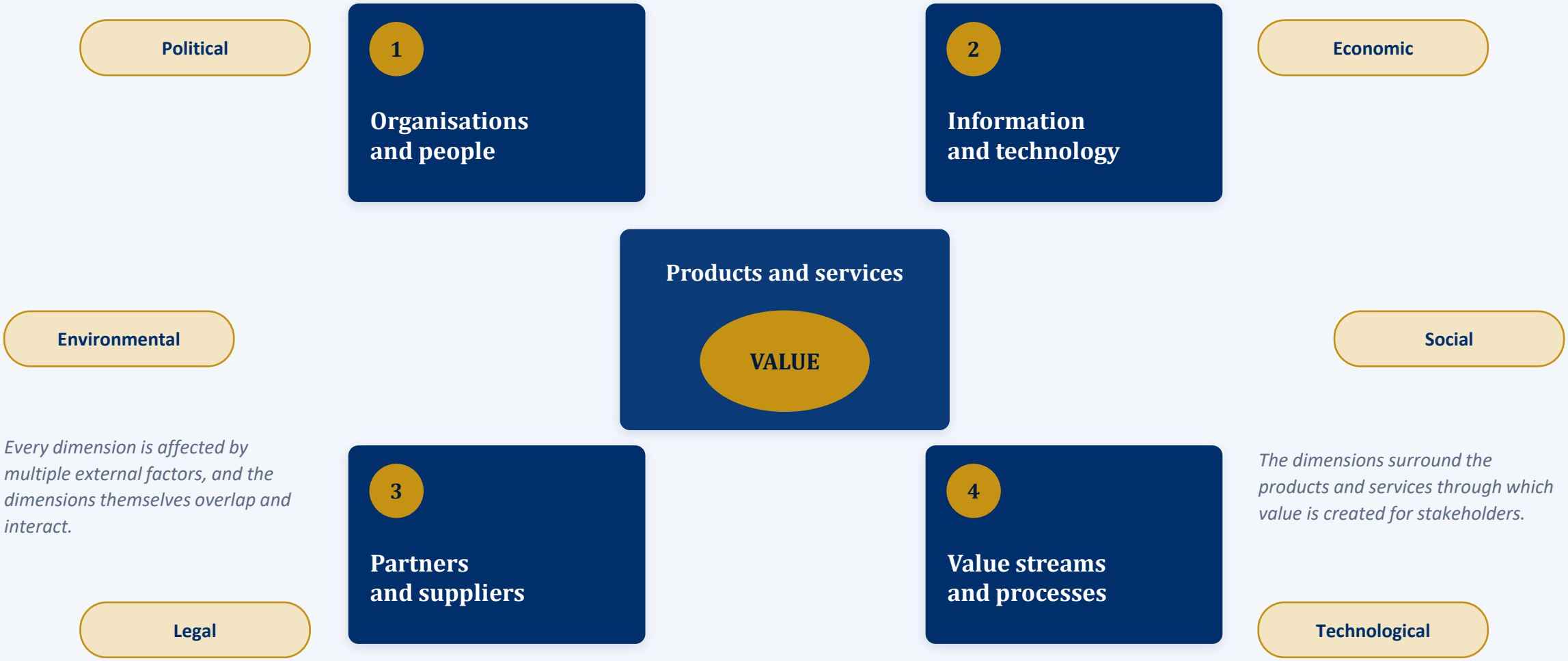
Relationships with other organisations, together with contracts and other agreements.

4 Value streams and processes

How activities, workflows, controls and procedures are organised to deliver agreed objectives.

ADAPTED FROM FIGURE 3.1

The Four Dimensions of Service Management



CONSEQUENCES

What Happens When a Dimension Is Neglected

1	**Organisations and people**	Skills gaps, resistance to change, silo behaviour and services that nobody is accountable for.
2	**Information and technology**	Poor data quality, incompatible systems, security exposure and technology that cannot be maintained.
3	**Partners and suppliers**	Outsourced services drift out of alignment with what the organisation actually needs.
4	**Value streams and processes**	Wasteful work, duplicated effort, and activity that conflicts with work being done elsewhere.

Failure to address all four dimensions can leave a service undeliverable, or unable to meet expectations of quality and efficiency.

BOUNDARIES ARE NOT SHARP

The Dimensions Overlap and Interact

No clean lines

The four dimensions do not have sharp boundaries. They may overlap, and they sometimes interact in ways that cannot be predicted in advance, depending on the level of complexity and uncertainty in which the organisation operates.

Where they overlap

A service desk agent (people) uses a knowledge base (technology) supplied by a vendor (partner) while following a procedure (process). One activity touches all four dimensions at once.

Why prediction fails

Changing one dimension shifts the others. Automating a workflow changes required competencies, alters supplier scope and redefines the process itself.

The practical rule

Consider all four dimensions for every service, and address each of them when managing and improving the SVS at every level of the organisation.

1

Organisations and People

Structures, roles, systems of authority, culture, capacity and competence

DIMENSION 1

Organisations and People: Scope

KEY MESSAGE

As organisations grow more complex, it becomes important that the way an organisation is structured and managed, together with its roles, responsibilities and systems of authority and communication, is well defined and supports the overall strategy and operating model.

1 Roles and responsibilities

Who does what, and who is accountable for the outcome.

2 Formal structures

Reporting lines, teams, departments and systems of authority.

3 Culture

The shared values and attitudes that emerge from how work is really carried out.

4 Staffing and competencies

Having enough people, with the right skills, in the right places.

DIMENSION 1

Structure Alone Does Not Deliver Effectiveness

1 A formal structure is necessary but insufficient

An organisation chart and a system of authority cannot by themselves guarantee that an organisation will be effective.

2 Culture must support the objectives

The organisation also needs a culture aligned to its objectives, and adequate capacity and competency in its workforce.

3 Leaders must champion values

Leaders are expected to advocate the values that motivate people to work in desirable ways, and to model them consistently.

4 But behaviour writes the culture

Ultimately it is the way the organisation carries out its work that creates the shared values and attitudes which, over time, become its culture.

DIMENSION 1

Building a Culture of Trust and Transparency

Why trust matters

A culture of trust and transparency encourages members of an organisation to raise and escalate issues early. Corrective action can then be taken before customers ever feel the impact.

Psychological safety

Staff report problems without fear of blame. Incidents surface while they are still small and cheap to fix.

Early escalation

Escalation is treated as responsible professional behaviour, not as an admission of personal failure.

Start with the guiding principles

Adopting the ITIL guiding principles offers a practical starting point for establishing a healthy organisational culture.

DIMENSION 1

People, Skills and Competencies

WHO COUNTS AS PEOPLE HERE?

Customers, employees of suppliers, employees of the service provider, and any other stakeholder in the service relationship. Everyone in the value network belongs to this dimension.

Beyond technical skill

Attention is due not only to the skills of teams and individuals, but also to management and leadership styles, and to communication and collaboration skills.

Skills must keep evolving

As practices evolve, people must continually update their skills and competencies. A qualification obtained once does not remain current.

Understanding the interfaces

People increasingly need to understand where their role connects to the roles of others, so that collaboration and coordination actually happen.

Broad knowledge, deep specialism

In software development and user support especially, the expectation is a broad general knowledge of the organisation combined with deep expertise in a chosen field.

DIMENSION 1

Breaking Down Organisational Silos

Every person in the organisation should be able to explain their own contribution to creating value for the organisation, its customers and other stakeholders.

A The silo symptom

Teams optimise their own metrics, hand work over the wall, and treat the next team as the customer rather than the end user.

B The remedy

Promoting a shared focus on value creation is an effective method of breaking down silos, because it gives every team a common reference point.

C In the Ghanaian context

Consider a university that separates admissions, finance and IT. A single applicant experiences one service, not three departments.

ILLUSTRATIONS

Applying Dimension 1

Axle Car Hire (the ITIL running case)

For Axle's car hire service, the organisations and people dimension covers far more than the IT team. It includes procurement, human resources, facilities and every other team whose work shapes the customer's experience of hiring a car.

A Ghanaian parallel: a mobile money service

The people dimension covers the agents in the field, the contact centre staff, the risk and compliance officers, the regulator's supervisors, and the merchants who accept payment. A control designed only for head office staff will fail at the agent point.

2

Information and Technology

The information and knowledge services need, and the technologies that enable them

DIMENSION 2

Information and Technology: Scope

KEY MESSAGE

Applied to the SVS, this dimension covers the information and knowledge needed to manage services and the technologies required to do so. It also covers the relationships between different components of the SVS, such as the inputs and outputs of activities and practices.

1 Applied to service management

The tools and knowledge the provider uses to run services: ticketing, monitoring, configuration data, reporting.

2 Applied to the service itself

The information created, managed and used during service provision and consumption, and the technology that enables it.

3 Applied to the connections

How components of the SVS exchange information, including the inputs and outputs of activities and practices.

DIMENSION 2

Technologies That Support Service Management

1 Workflow management systems

Route, track and control work items end to end.

2 Knowledge bases

Capture and reuse solutions, reducing repeat effort.

3 Inventory systems

Record what assets exist, where they are and who holds them.

4 Communication systems

Connect users, teams and suppliers across channels.

5 Analytical tools

Turn operational data into decisions and forecasts.

6 AI and machine learning

Applied from strategic planning and portfolio optimisation through to system monitoring and user support.

Mobile platforms, cloud solutions, remote collaboration tools, automated testing and deployment solutions have all become common practice among service providers.

DIMENSION 2

In the Context of a Specific IT Service

1 It covers the whole architecture

The technologies involved usually span every level of IT architecture: applications, databases, communication systems and the integrations between them.

2 It depends on the service

The specific information and technologies required depend entirely on the nature of the service being provided.

3 Emerging technology as differentiator

Blockchain, artificial intelligence and cognitive computing can give early adopters a business differentiation advantage, particularly in highly competitive industries.

4 Yesterday's edge is today's baseline

Cloud computing and mobile applications were once differentiators. They have become common practice across many industries globally.

DIMENSION 2

Three Questions About the Information Component

?

What information is managed by the services?

Identify the data the service holds, creates and transmits, and who owns it.

?

What supporting information and knowledge are needed to deliver and manage the services?

Consider runbooks, configuration records, service level targets and staff know-how.

?

How will information and knowledge assets be protected, managed, archived and disposed of?

Address the full lifecycle, not only the point of collection.

DIMENSION 2

Information Is Often the Real Product

THE CENTRAL POINT

For many services, information management is the primary means of enabling customer value. Information is generally the key output of the majority of IT services consumed by business customers.

An HR service

It creates value by letting the organisation access and maintain accurate information about employees, their employment and their benefits, while keeping private information away from unauthorised parties.

A network management service

It creates value by maintaining accurate information about active network connections and utilisation, so that bandwidth capacity can be adjusted intelligently.

DIMENSION 2

Information Architecture: Six Quality Criteria

How information is exchanged between services and service components is a key consideration. The information architecture must be well understood and continually optimised against the following criteria:

1 Availability

The information is there when it is needed.

2 Reliability

It can be trusted to be consistent and complete.

3 Accessibility

The right people can reach it through the right channel.

4 Timeliness

It arrives early enough to inform the decision.

5 Accuracy

It correctly represents the real world.

6 Relevance

It answers the question actually being asked.

DIMENSION 2

Security, Regulation and Compliance

A defining constraint

The challenges of information management created by security and regulatory compliance requirements are a central focus of this dimension. Regulation shapes what an organisation may collect, where it may store it, and how long it may keep it.

European Union: GDPR

The General Data Protection Regulation shapes information management policies and practices for any organisation within its reach, including many Ghanaian firms serving European clients.

United States: HIPAA

The Health Insurance Portability and Accountability Act of 1996 sets data privacy and security provisions for safeguarding medical information.

Ghana: Act 843 and Act 1038

The Data Protection Act, 2012 requires registration of data controllers and lawful processing. The Cybersecurity Act, 2020 introduces obligations for critical information infrastructure owners.

Choosing a Technology: Eight Questions (1 of 2)

When considering a technology for the planning, design, transition or operation of a product or service, an organisation should ask:

1 Compatibility

Does it fit the current architecture of the organisation and its customers? Do the different products used by the organisation and its stakeholders actually work together?

2 Disruption risk

How are emerging technologies such as machine learning, artificial intelligence and the Internet of Things likely to disrupt this service or the organisation?

3 Compliance

Does it raise regulatory or other compliance issues with the organisation's policies and information security controls, or with those of its customers?

4 Viability

Will it remain viable in the foreseeable future? Is the organisation willing to accept the risk of ageing technology, or of unproven emerging technology?

Choosing a Technology: Eight Questions (2 of 2)

5 Strategic alignment

Does the technology align with the strategy of the service provider, or with that of its service consumers?

6 Skills

Does the organisation have the right skills across its own staff and its suppliers to support and maintain the technology over its life?

7 Automation capability

Does it offer enough automation capability to be efficiently developed, deployed and operated?

8 Leverage and risk

Does it offer additional capabilities that could serve other products or services? And does it introduce new risks or constraints, such as locking the organisation into a specific vendor?

DIMENSION 2

Culture and Industry Shape Technology Choice

Organisational culture

Some organisations want to be at the cutting edge of technological advance. Others are more traditional by disposition. One company may be eager to exploit artificial intelligence while another is barely ready for advanced data analysis tools. Long-standing preference is difficult to shift without a compelling reason.

The nature of the business

A company doing significant business with government clients may face restrictions on certain technologies, or much higher security requirements. Finance and life sciences face similar constraints, and often cannot use open source or public services when handling sensitive data.

Cloud Computing: Definition and Characteristics

DEFINITION

Cloud computing is a model for enabling on-demand network access to a shared pool of configurable computing resources that can be rapidly provided with minimal management effort or provider interaction.

1 On-demand availability

Often self-service, without human intervention.

2 Network access

Typically over the public internet.

3 Resource pooling

Often shared among multiple organisations.

4 Rapid elasticity

Often automatic, in both directions.

5 Measured service

Usually from the consumer's own perspective.

How Cloud Adoption Changes the In-House Provider

Infrastructure	Some infrastructure previously managed in house is replaced by a partner's cloud service.
Expertise	The need for infrastructure management expertise and provider-owned resources decreases or disappears.
Monitoring and control	Focus shifts from the in-house infrastructure to a partner's services.
Cost structure	Specific capital expenditure is removed and new operating expenditure appears, which must then be managed.
Network and security	Requirements for network availability and security rise sharply.
Risk and compliance	New security and compliance risks apply to both the provider and the partner delivering the cloud service.
Consumption	Users can scale consumption themselves through simple standard requests, or with no request at all.

Practices Affected by the Cloud Model

1. Service level management
2. Measurement and reporting
3. Information security management
4. Service continuity management
5. Supplier management
6. Incident management
7. Problem management
8. Service request management
9. Service configuration management

Adoption of cloud computing changes service architecture and redistributes responsibility between consumers, providers and partners. Its effects are felt across many practices at once.

Cloud, Speed and the Strategic Decision

1 Elasticity enables speed

Because computing resources are elastic, cloud infrastructure can support significantly faster deployment of new and changed services, supporting high-velocity service delivery.

2 A prerequisite for DevOps

The ability to configure and deploy computing resources as quickly as new applications are written is an important precondition for the success of DevOps and similar initiatives.

3 It answers a business need

This supports the modern organisation's need for faster time to market and for the digitalisation of its services.

4 Decide it at strategic level

Decisions about the use of the cloud model belong at the strategic level of the organisation and should involve stakeholders at every level, from governance through to operations.

3

Partners and Suppliers

Relationships with other organisations, and the contracts and agreements that govern them

DIMENSION 3

Partners and Suppliers: Scope

KEY MESSAGE

This dimension encompasses an organisation's relationships with other organisations involved in the design, development, deployment, delivery, support or continual improvement of services. It also covers the contracts and other agreements between them.

1 Nobody stands alone

Every organisation, and every service, depends to some extent on services provided by other organisations.

2 Varying integration

Relationships range from formal contracts with a clear separation of responsibilities to flexible partnerships built on shared goals and shared risk.

3 A spectrum, not a set of boxes

The forms of cooperation are not fixed. An organisation's position on the spectrum varies with its strategy and its objectives for the relationship.

ADAPTED FROM TABLE 3.1

Relationships Between Organisations

Form of cooperation	Outputs	Responsibility for outputs	Responsibility for outcomes	Level of formality	Example
Goods supply	Goods supplied	Supplier	Customer	Formal supply contract and invoices	Procurement of computers and phones
Service delivery	Services delivered	Provider	Customer	Formal agreements with flexible cases	Cloud computing: infrastructure or platform as a service
Service partnership	Value co-created	Shared between provider and customer	Shared between provider and customer	Shared goals, generic agreements, case-based arrangements	Employee onboarding shared between HR, facilities and IT

As cooperation moves down the table, responsibility for outcomes shifts from resting with the customer alone to being genuinely shared.

DIMENSION 3

The Sourcing Continuum

When an organisation acts as a service consumer, the role it takes depends on its strategy and objectives for sourcing and supplier management. That strategy should be grounded in the organisation's goals, culture and business environment.



Retain and control

Some organisations choose to rely as much as possible on their own resources, using partners and suppliers as little as they can, in order to retain full control over important functions.

The many variations between

In practice most organisations sit somewhere between the two poles, and their position shifts over time as cost, capability and risk appetite change.

Focus and delegate

Others believe they are best served by developing certain core competencies themselves and relying on partners and suppliers to meet the remaining needs.

DIMENSION 3

Service Integration and Management

One way to address this dimension

Service integration and management involves the use of a specially established integrator whose task is to ensure that service relationships across many suppliers are properly coordinated.

Keep it in house

The organisation builds its own integrator function, retaining control over supplier coordination and holding the relationship knowledge internally.

Delegate it to a trusted partner

The integration role is contracted to a partner. This demands considerable trust, since that partner then coordinates the work of its own competitors.

Why it matters

Where a single service depends on five or six suppliers, no individual contract can guarantee the end-to-end outcome. Somebody must own the seams between them.

Factors Influencing Supplier Strategy (1 of 2)

1 Strategic focus

Some organisations prefer to concentrate on their core competency and outsource non-core supporting functions to third parties. Others prefer to remain as self-sufficient as possible and retain full control.

2 Corporate culture

Some organisations have a historical preference for one approach over another. Long-standing cultural bias is difficult to change without a compelling reason to do so.

3 Resource scarcity

Where a required resource or skillset is in short supply, the service provider may find it difficult to acquire what it needs without engaging a supplier.

4 Cost concerns

The decision may turn on whether the provider believes it is more economical to source a particular requirement from a supplier than to build it internally.

Factors Influencing Supplier Strategy (2 of 2)

5 Subject matter expertise

The provider may judge it less risky to use a supplier that already holds expertise in a required area than to try to develop and then maintain that expertise in house.

6 External constraints

Government regulation or policy, industry codes of conduct, and social, political or legal constraints all shape an organisation's supplier strategy.

7 Demand patterns

Customer activity may be seasonal or highly variable. Such patterns influence how far an organisation uses external providers to absorb variable demand.

In Ghanaian practice

The Public Procurement Act, 2003 (Act 663) as amended, and the thresholds and approval routes it prescribes, are a live example of external constraint on supplier strategy.

DIMENSION 3

The Rise of 'As a Service'

A DECADE OF CHANGE

The last decade has seen an explosion of companies offering technical resources and capabilities 'as a service'. They bundle goods and services into a single product offering that is consumed as a utility.

1 Infrastructure as a service

Technical resources such as compute, storage and networking, consumed on demand.

2 Platform and software as a service

Capabilities rather than raw resources: development platforms and complete applications.

3 From capital to operating expenditure

This frees organisations from investing in costly infrastructure and software assets that would be accounted for as capital expenditure.

4

Value Streams and Processes

How the parts of the organisation work together, in an integrated and coordinated way

DIMENSION 4

Value Streams and Processes: Scope

KEY MESSAGE

Applied to the organisation and its SVS, this dimension is concerned with how the various parts of the organisation work in an integrated and coordinated way to enable value creation through products and services.

1 What activities are undertaken

The dimension focuses on the work the organisation actually does, and on how that work is organised.

2 How value creation is assured

It also covers how the organisation satisfies itself that it is enabling value creation for all stakeholders both efficiently and effectively.

3 What it defines

In both contexts it defines the activities, workflows, controls and procedures needed to achieve agreed objectives.

From the Service Value Chain to Value Streams

1

Operating model

ITIL gives service providers an operating model covering all the key activities needed to manage products and services effectively.

2

The service value chain

That model is the ITIL service value chain, examined in detail in section 4.5 of the syllabus.

3

Generic by design

The value chain operating model is deliberately generic. In practice it can be followed in many different patterns.

4

Value streams

Those patterns within the operation of the value chain are what ITIL calls value streams.

Definition: Value Stream

DEFINITION

A value stream is a series of steps an organisation undertakes to create and deliver products and services to consumers.

It is a combination of value chain activities

A value stream draws on the organisation's value chain activities and arranges them into a specific sequence for a specific purpose. Appendix A of the syllabus provides worked examples.

It is defined per product or service

Value streams should be defined by organisations for each of their products and services, rather than assumed to be identical across the portfolio.

It may be stable or fluid

Depending on strategy, value streams may be redefined to react to changing demand and other circumstances, or may remain stable for a considerable period.

Value Stream Mapping and the Elimination of Waste

Why map at all?

Identifying and understanding the value streams an organisation has is critical to improving overall performance. Structuring activities as value streams gives a clear picture of what is delivered and how, and makes continual improvement possible.

1 Map what actually happens

Organisations should examine how they really perform work and map every value stream they can identify, not the idealised version in the manual.

2 Analyse and find the barriers

Mapping enables analysis of the current state and identification of barriers to workflow and of non-value-adding activity, that is to say, waste. Wasteful activity should be eliminated to raise productivity.

3 Then add value

Opportunities to increase value-adding activity exist across the value chain. Optimisation may include process automation or the adoption of emerging technologies and new ways of working to gain efficiency or improve user experience.

DIMENSION 4

Processes, Procedures and Work Instructions

DEFINITION

A process is a set of interrelated or interacting activities that transform one or more defined inputs into defined outputs. Processes define the sequence of actions and their dependencies.

1 Process

Describes what is done to accomplish an objective. Well-defined processes can improve productivity within and across organisations.

2 Procedure

Outlines who is involved in the process and in what capacity. Processes are usually detailed in procedures.

3 Work instruction

Explains how the steps are actually carried out, in enough detail for the work to be performed consistently.

DIMENSION 4

Value Stream Compared with Process

Point of comparison	Value stream	Process
Primary focus	The end-to-end journey that delivers something to a consumer	The transformation of defined inputs into defined outputs
Scope	Crosses teams, practices and often organisations	Usually contained within a defined area of responsibility
Built from	A combination of value chain activities	A set of interrelated or interacting activities
Question answered	How does value reach the consumer, and where is it delayed?	What is done, in what sequence, and with what dependencies?
Detailed in	Value stream maps for each product or service	Procedures and work instructions

DIMENSION 4

Three Questions This Dimension Answers

1

What is the generic delivery model for the service, and how does the service work?

Establishes the shape of delivery before any detail is designed.

2

What are the value streams involved in delivering the agreed outputs of the service?

Identifies the end-to-end sequences that must be mapped and improved.

3

Who, or what, performs the required service actions?

Note the 'or what': automation and machines are legitimate performers of service actions.

The specific answers will vary depending on the nature and the architecture of the service.

5

External Factors

The PESTLE forces that constrain and influence every service provider

EXTERNAL FACTORS

The PESTLE Model

Service providers do not operate in isolation. They work in dynamic and complex environments that can show high volatility and uncertainty, and that impose real constraints on how the provider can work.

P Political

Government stability, policy direction, public sector priorities and political interference in delivery.

E Economic

Exchange rates, inflation, interest rates, purchasing power and the cost of capital.

S Social

Demographics, literacy, digital inclusion, customer expectations and societal attitudes.

T Technological

Rate of technological change, infrastructure availability and the emergence of new platforms.

L Legal

Statutes, regulation, licensing regimes, contracts and data protection obligations.

E Environmental

Climate, energy, sustainability expectations and environmental reporting requirements.

EXTERNAL FACTORS

How External Factors Reshape the Dimensions

1 Environmental expectations

Government and societal attitudes towards environmentally friendly products may push an organisation to invest more in tools and technologies that meet those expectations, and to partner with providers who can demonstrate green credentials. Some companies now publish product environmental reports covering climate change and safer materials.

2 Economic and social segmentation

Economic and societal factors may lead organisations to create several versions of the same product for consumer groups with different buying patterns. Music and video streaming services illustrate this with a free advertising-supported tier, a premium tier without advertising, and family plans covering several profiles under one account.

3 Data protection law

Data protection laws and regulations have changed how companies collect, process, access and store customer data, and how they work with external partners and suppliers. The obligation flows down the supply chain, not only to the organisation itself.

WORKED EXAMPLE

PESTLE Applied: A Ghanaian Digital Service

Factor	Illustrative pressure on a Ghanaian service provider	Dimension most affected
Political	Shifting digitalisation policy priorities and changes in ministerial direction between administrations	Partners and suppliers
Economic	Currency depreciation raising the cedi cost of dollar-denominated cloud subscriptions and licences	Information and technology
Social	Uneven digital literacy and continuing reliance on assisted service channels in rural districts	Organisations and people
Technological	Expanding fibre and 4G coverage alongside persistent power reliability challenges	Value streams and processes
Legal	Data Protection Act, 2012 (Act 843) and Cybersecurity Act, 2020 (Act 1038) compliance obligations	Information and technology
Environmental	Electronic waste management expectations and pressure for energy-efficient data centre choices	Partners and suppliers

6

Contemporary Issues

What is changing in each dimension right now

CONTEMPORARY ISSUE 1

Generative AI and the Four Dimensions

1	**Organisations and people**	Prompt literacy becomes a competency. Roles shift from executing routine tasks to reviewing machine output. Fear of displacement is a genuine cultural risk.
2	**Information and technology**	Model choice, data residency, retrieval architecture and hallucination controls become architectural decisions rather than experiments.
3	**Partners and suppliers**	Dependence on a small number of model providers concentrates risk. Contracts must address training data use, retention and portability.
4	**Value streams and processes**	Automation of triage, drafting and knowledge retrieval removes steps from value streams. The remaining human steps must be redesigned, not merely preserved.

CONTEMPORARY ISSUE 2

Data Sovereignty and Cloud Localisation

The tension

Cloud economics push workloads towards a handful of global regions. Law, policy and public expectation push sensitive data towards local storage. Service providers must resolve the two.

Regulatory pressure

Data protection regimes increasingly restrict cross-border transfer of personal data, or require specific safeguards and approvals before it occurs.

Architectural consequence

Providers adopt hybrid patterns: sensitive records held locally, elastic compute drawn from global regions, with careful control of what crosses the boundary.

Supplier consequence

Contracts must specify processing location, sub-processor disclosure, breach notification timelines and exit provisions. This is a partners and suppliers issue as much as a technology one.

CONTEMPORARY ISSUE 3

Cyber Resilience and Third-Party Risk

1 The attack surface has moved outwards

Most serious incidents now involve an identity, a supplier or an internet-facing service rather than a breach of the corporate perimeter.

2 Supplier concentration is systemic risk

When many organisations depend on the same platform, a single supplier failure becomes a sector-wide outage. Concentration must be assessed, not merely noted.

3 Resilience is a people property

Detection and response depend on trained staff, rehearsed procedures and a culture where bad news travels fast and without blame.

4 Regulatory expectation is rising

Critical information infrastructure designations, incident reporting duties and audit obligations now shape design decisions from the outset.

CONTEMPORARY ISSUE 4

Skills, Ways of Working and Sustainability

1 The skills equation

Demand for cloud, data and security competence outstrips local supply. Organisations respond by growing talent internally, by partnering, or by accepting delivery risk. Each choice sits in a different dimension.

2 Product over project

Long-lived product teams that own a service end to end are displacing temporary project structures. This changes reporting lines, funding models and the shape of value streams simultaneously.

3 Sustainable IT

Energy consumption, hardware lifecycle and electronic waste have become reportable concerns. Sustainability now enters technology selection alongside cost and capability.

APPLICATION

Integrated Case: A National Digital Payment Service

Dimension	What sits inside it	A failure symptom if neglected
Organisations and people	Agents, contact centre staff, compliance officers, merchant support teams, the regulator's supervisors	Fraud losses at the agent point despite sound central controls
Information and technology	Core switch, ledgers, KYC records, fraud analytics, settlement interfaces, mobile applications	Reconciliation breaks and customer disputes that cannot be resolved
Partners and suppliers	Telecommunication operators, banks, the national switch, software vendors, cash-in-transit providers	An outage at one partner halts settlement for every participant
Value streams and processes	Onboarding, cash-in and cash-out, dispute resolution, settlement, agent liquidity management	Disputes take weeks because the value stream was never mapped end to end

TUTORIAL

Class Activity: The Four-Dimension Audit

Your brief

In groups of four, select one digital service you personally use in Ghana. Audit it across all four dimensions and present your findings in five minutes.

1 Identify and classify

List at least three concrete components of the service under each of the four dimensions.

2 Find the weakest dimension

Decide which dimension is least well served, and defend that judgement with observable evidence rather than opinion.

3 Trace the consequence

Explain how weakness in that dimension shows up as a problem the customer actually experiences.

4 Apply a PESTLE lens

Identify two external factors that constrain the provider, and state which dimension each one presses on hardest.

REVISION

Examination Guidance and Common Errors

1 Learn the four names exactly

Marks are lost for writing 'people' instead of 'organisations and people', or 'processes' instead of 'value streams and processes'.

2 Do not confuse dimensions with the SVS

The four dimensions apply to the SVS. They are not components of it. Chapter 4 sets out the components.

3 Keep value stream and process apart

A value stream is the end-to-end series of steps that delivers to a consumer. A process transforms defined inputs into defined outputs.

4 Remember PESTLE has six letters

Political, economic, social, technological, legal and environmental. Candidates frequently omit environmental.

5 Partners and suppliers is not only procurement

It includes contracts and agreements, but also partnerships where risk and goals are genuinely shared.

6 Cite the reason for holism

No single dimension is sufficient in isolation to produce the required outcomes. Say so explicitly in extended answers.

REVISION

Practice Questions

1. Which dimension covers an organisation's contracts and agreements with other organisations?

A. Organisations and people B. Information and technology C. Partners and suppliers D. Value streams and processes

2. Which statement about the four dimensions is correct?

A. They apply only to IT services B. They have sharp, fixed boundaries C. They apply to the SVS and to every service D. They replace the service value chain

3. What does the 'L' in PESTLE represent?

A. Logistical B. Legal C. Localisation D. Leadership

4. A series of steps an organisation undertakes to create and deliver products and services to consumers is a:

A. Process B. Procedure C. Practice D. Value stream

REVISION

Practice Questions: Answers and Rationale

1 — C **Partners and suppliers**

This dimension covers relationships with other organisations involved in the service, together with the contracts and other agreements between them.

2 — C **They apply to the SVS and to every service**

The dimensions apply to all services being managed and to the SVS in general. Their boundaries are not sharp and they may overlap.

3 — B **Legal**

PESTLE stands for political, economic, social, technological, legal and environmental factors.

4 — D **Value stream**

A process transforms defined inputs into defined outputs; a value stream is the series of steps that delivers to the consumer.

CONSOLIDATION

Chapter Summary

SUMMARY

The four dimensions represent a holistic approach to service management. Organisations should ensure a balance of focus between them, and should consider the impact of external factors on all four.

1 All four, all the time

Every dimension and every external factor should be addressed as they evolve, taking account of emerging trends and opportunities.

2 Consider the SVS from all four

It is essential that an organisation's SVS is examined from each of the four dimensions rather than from a favoured one.

3 The cost of imbalance

Failure to address or account for one dimension, or one external factor, leads to sub-optimal products and services.

4 Blend the ways of working

Effective organisations combine people, teams, value streams and ways of working, drawing on DevOps, design thinking, Agile and Lean, and automating where it is sensible to do so.

CONSOLIDATION

Five Things to Carry Away

1. Four dimensions, always together: organisations and people, information and technology, partners and suppliers, value streams and processes.
2. None of them is sufficient on its own to produce the outcomes an organisation requires.
3. They apply to the whole Service Value System and to every individual service, at every level.
4. Their boundaries are not sharp; they overlap and interact in ways that are not always predictable.
5. Six external PESTLE factors press on all four dimensions and lie largely beyond the organisation's control.

REFERENCE

Glossary of Key Terms

Value stream

A series of steps an organisation undertakes to create and deliver products and services to consumers.

Process

A set of interrelated activities that transform defined inputs into defined outputs, in a defined sequence.

Procedure

A documented outline of who is involved in a process and in what capacity.

Work instruction

A documented explanation of how the steps of a procedure are carried out.

Cloud computing

On-demand network access to a shared pool of configurable computing resources, rapidly provisioned with minimal management effort.

PESTLE

A model for analysing the political, economic, social, technological, legal and environmental factors affecting an organisation.

Service integration and management

The use of a specially established integrator to coordinate service relationships across multiple suppliers.

Culture

The shared values and attitudes that emerge over time from the way an organisation actually carries out its work.

REFERENCE

References and Further Reading

- AXELOS (2019) ITIL® Foundation: ITIL 4 Edition. London: TSO. Chapter 3, pp. 24–34.
- AXELOS (2019) ITIL® 4: Create, Deliver and Support. London: TSO. Extended guidance on value stream mapping.
- Republic of Ghana (2012) Data Protection Act, 2012 (Act 843).
- Republic of Ghana (2020) Cybersecurity Act, 2020 (Act 1038).
- Republic of Ghana (2003) Public Procurement Act, 2003 (Act 663), as amended by Act 914 (2016).
- Mell, P. and Grance, T. (2011) The NIST Definition of Cloud Computing. NIST Special Publication 800-145.

- ITIL® is a registered trade mark of AXELOS Limited. These slides are a teaching summary prepared for UPSA students and are not a substitute for the official publication.

Questions and Discussion

Next session: Chapter 4 — The ITIL Service Value System, the guiding principles, governance and the service value chain.

Prepare by reading Chapter 4, sections 4.1 to 4.3, and by completing the four-dimension audit for your chosen service.

University of Professional Studies, Accra • Scholarship with Professionalism